

Investigating Multimodal Learning with Inconsistent Modalities and Attention-Enhanced Approaches

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ABSTRACT

Human activity recognition (HAR) is one of the most important and challenging problems in the computer vision space. Applications include tasks such as gaming, human-robot interaction, rehabilitation, sports, health monitoring, video surveillance, and robotics. Sensors for multiple modalities could be used to overcome some of these inherent challenges. Such sensors could include an RGB-D camera, infrared sensors, and inertial sensors. Multimodal learning enables networks to consider multiple sensors when performing activity recognition. This paper proposes networks which use attention to better focus on select portions of data along both embedding-space and time. Projection networks, which project from one modality to another in order to use a multimodal network, are also proposed as an alternative when the second modality is unavailable. It is observed that adding attention improves performance and that using projected data retains most of the performance from multimodal architectures.

KEYWORDS

activity recognition, video classification, RGB-D, , neural networks, data hallucination, generative systems, multi-modal learning

1 INTRODUCTION

Activity recognition aims to recognize actions of an agent based on a set of observations. The space of activity recognition has grown considerably over the past decade. From smart watches to smart security, these systems and devices are able to accurately detect actions from cameras and a plethora of sensors such as accelerometers. Additionally, there is plenty of data available: IMU (Inertial Measurement Unit) data, skeleton data, accelerometer data, and video data [5]. With these advancements, the space of multi-modal learning and the ability to classify actions with multiple data sources has advanced immensely. With the rise of multiple sources of sensor and video data, feature extraction and fusion of these features for multi-modal learning has become a larger part of deep learning research. In terms of feature extraction, one of the most recent advancements has been turning numerical sensor data into an image (an image representing one data point with data from multiple sensors) and inputting it into a convolutional neural network[5]. The technique has been widely regarded as one of the top performers when it comes to classification accuracy. Additionally, the 3D convolutional neural network has become an important advancement for feature extraction of video data [3]. Lastly, with this ability to feature extract, there are architectures which now support multi-modal learning (including both video and sensor data) and boast high accuracies after feature extraction and fusion [10].

Attention-enhanced approaches are recently being popular approaches to enhance models. Attention allows for the model to focus

on the most important parts of the input. With techniques from Vaswani et al. [11] and Murahari et al. [6], attention can be applied to make sure the most important part of the input is considered when running the model.

Along with scarce data, there is often a missed modal input for the current architecture/network in place. To overcome missing inputs, we can use hallucination techniques, such as generative systems, where the system will generate real data. Primarily, work has been completed in cross-modality hallucination, which includes training a system to turn text into image data. Cross-modality hallucination has proven to be successful and can be used to fill datasets where a certain part of the data is incomplete[8]. The goal of our research is to determine whether it is possible to build a multi-input network that trains a model across many sensor data types and is capable of training on datasets that have only one or a few of the input modalities. Additionally, we look into hallucination and attention-enhanced techniques to see if they provide a benefit to multi-modal learning. The next section will give a glimpse into the recent state of research in the fields of multi-modal learning, attention techniques, and modality hallucination and provide insight into what next steps might be. The dataset we will focus on using is the 50salads dataset.

2 LITERATURE REVIEW

In the past decade, the space of activity recognition has become increasingly advanced. With more evolved sensor capability and continuously better GPU performance, there have been strides in the multi-modal learning space to learn and classify actions with data from multiple sensors. In many activity recognition tasks, we have sensor modalities such as pose data, RGB, and accelerometer. However, there are few datasets that have all of these modalities. With the set up described previously, is it possible to build a multi-input dataset that trains a representation across many modalities, and is capable of training on datasets that have only one or a few of the input modalities? In this literature review, we will be able to analyze the current research in the space of multi-modal learning and get a chance to determine the next steps for the research area of interest.

2.1 Feature Extraction and Multi-Modal Learning

The area of multi-modal learning has expanded as sensors capabilities have improved. Feature extraction research has grown substantially. Memmesheimer et al. research into feature extraction provides evidence that encoding sensor data into image data is effective in classification[5]. In his approach, each vector represents a discrete 1-dimensional signal, and each column vector represents a sample of all sensors at one specific time step. Once the data is encoded into image data, it is much easier to feature extract and

classify the action data using convolutional neural networks. In fact, the encoding sensor data approach does better than most multi-modal learning architectures, achieving a 93.33 % accuracy rate on the UTD-MHAD dataset [5]. Overall, this approach is effective with data such as IMU, inertial, and skeleton data. However, the encoding sensor data approach does not hold when video data is an input into our model. There must be a way to extract and process video data. The usage of a 3D convolutional neural network will fit the problem of activity recognition in video. The 3D CNN model extracts 4 features from both spatial and temporal dimensions by performing 3D convolutions, thereby capturing the motion information encoded in multiple adjacent frames. In Ji et al. 's research on 3D CNNs, precision is higher than a rolling 2D convolutional neural network and is an effective method to extract data from video to classify actions [3]. Since our research involves multi-modal learning and using inputs from multiple sensors, an architecture similar to Tao's for multi-modal learning would be an effective architecture[10]. With multiple modalities, we can first feature extract with CNNs with sensor modalities such as skeleton and feature extract with 3D CNNs with video modality, which incorporates both approaches of Memmesheimer et. al 's and Ji et. al 's feature extraction and activity recognition research. After the feature extraction is complete, we can fuse the extract features and classify. Tao et al 's research, with his feature extraction methods and sensor fusion methods, boasts an accuracy of above 94 %. With this feature extraction and sensor fusion research from three papers, we will run experiments to see which one would fare better, including considering our other constraints.

2.2 Attention-Enhanced Approaches

Attention mechanisms provide an ability to utilize the most relevant parts of the input sequence in a flexible manner, by a weighted combination of all of the encoded input vectors, with the most relevant vectors being attributed the highest weights. For instance, assume a sequence of words is given. An attention-enhanced analysis approach takes the query vector attributed to some specific word in the sequence and scores it. In doing so, it captures how the word under consideration relates to the others in the sequence. Then it scales the values according to the attention weights computed from these scores, in order to retain focus on those words that are most relevant. In doing so, it produces an attention output for the sequence of words under consideration. Attention mechanisms are a relatively new to the activity recognition world. Vaswani et. al highlights proposes a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely [11]. Additionally, Murahari et. al. highlights the use of attention in activity recognition, adjustments to DeepConvLSTM with an attention layer to output an embedding to pass into the model [6].

2.3 Generative Systems: Hallucinating Data

During experimentation in my research, one or more modalities may be missing. In this case, we will be experimenting data for a specific modality depending on which modality is missing. Pahde's research in cross-modal data hallucination builds a generative system by training on image and text data to classify on image data

[8]. Data hallucination is effective, proving to be a viable option for experimentation. From this point, there are various generative systems, but the problem to solve lies here. In our experimentation, we will try to hallucinate data with cross modal techniques such as passing in and try to pass in random data to see if improvements will be made in the activity recognition model's accuracy.

3 METHODOLOGY

From the 50salads dataset, we will be using the RGB and IMU data. The RGB and IMU data will be transformed such that it can be inputted into the models. Then, single and multi modal approaches will be used to predict the action being done. IMU is accelerometer data gathered from devices which measure acceleration along three-axes. We have access to ten devices attached to ten different utensils which produce IMU data, so our data is represented as a vector of 30 dimensions. RGB-D sensors are a specific type of depth-sensing devices that work in association with a RGB (red, green and blue color) sensor camera. They are able to augment the conventional image with depth information on a per pixel basis, but only the RGB portion will be used in experimentation.

3.1 Data Acquisition and Preprocessing

Our experiments start with data acquisition, where we have decided to use the 50salads dataset [1]. This dataset captures 25 people preparing 2 mixed salads each and contains over 4h of annotated accelerometer and RGB-D video data. The 50 Salads dataset has been used in research areas such as activity recognition, activity spotting, sequence analysis, progress tracking, sensor fusion, transfer learning, and user-adaptation. In our experiments, we will be using the accelerometer (IMU) data and RGB data as our two modalities for our modeling. Next, we need to preprocess the data to enable us to use in our multi-modal models. For the RGB data, Fraha [2]uses an anticipation framework to generate embeddings for the RGB data. This includes using a recurrent encoder, where observed frames in a single vector will be used to decode the future activities. Given the input features X, the recurrent encoder passes these features through a single layer with a gated recurrent unit. Each video will result in 2048 features for the n number of timestamps it has (n is dependent on video length). We will concatenate these videos into one large array with dimensions of m x 2048, m being the total number of timestamps for all videos combined. For the IMU data, accelerometer data is already given in vector of length 30 per data point.

3.2 RGB Modality

3.2.1 Multi Layer Perceptron. Our first experiment begins with a dense neural network, which classifies the RGB embeddings that were pre-processed from the 50 salads dataset through a linear layer. With this simplistic approach, training times will be faster and will be used as a baseline to assess more complicated networks. A downside of a simple network is the limited functionality, as a multilayered perceptron will poorly capture the features and traits of the IMU data.

3.2.2 MS-TCN++: Multi-Stage Temporal Convolutional Network for Action Segmentation. Our second experiment with RGB data will

use MS-TCN++ model, a multi-stage temporal convolutional network for action segmentation. The first stage of the model generates an initial prediction that is refined by the next ones. In each stage, several layers of dilated temporal convolutions are stacked, covering a large receptive field. While this architecture already performs well, lower layers still suffer from a small receptive field. To address this limitation, a dual dilated layer that combines both large and small receptive fields is used. This model is able to collect contextual information and segments the videos into actions. This model will take the RGB embeddings as input and output the probability of each action for inputted RGB embedding [4].

3.3 IMU Modality

3.3.1 Multi Layer Perceptron. The IMU network input is a vector of 30 dimensions and outputs a probability for each possible class the vector could be representing. The network is composed of dense linear layers with dropout, batchnorm, and non-linearity (ReLU) functions in the hidden layers. Similar to the simplistic RGB model, training times will be faster and will be used as a baseline to assess more complicated networks. A downside of the network is its inability to accurately capture the features and traits of the IMU data.

3.3.2 Deep Convolutional and LSTM Network. The DeepConvLSTM network proposed by Ordóñez and Roggen aims to improve upon the downsides of the simple IMU network from above. [7]. The network is composed of convolutional and LSTM recurrent layers, and this architecture is capable of automatically learning feature representations. This means the architecture will be able to view IMU data and interpret it as IMU data instead of a series of 30 independent points.

The convolutional layers within DeepConvLSTM take as input the IMU data as a set of 50x30 images. The 50 IMU data points in each step means the convolutional kernels capture information about the IMU sensor data over time. At the end of the convolutional layers, the network has computed an embedding for the set of 50 IMU data points. This embedding is passed into a set of LSTM layers which use with hidden states. The hidden states are computed from previous IMU steps, which means the LSTMs allows the network to learn over time beyond individual steps.

3.4 Combining RGB and IMU Modalities

3.4.1 Feature Concatenation and Multi Layer Perceptron. First, we must build an initial multi-modal model where we input 2 modalities, IMU and RGB data from the 50salads dataset, and output a prediction of the activity taking place.

In our data acquisition and preprocessing steps, we used a CNN to generate embeddings/features from the RGB video data and used an IMU network to generate embeddings for the accelerometer data. Then, the two embeddings generated will be concatenated into a single feature set and will be passed into the MLP. With more data, the hope is to get better predictions.

3.5 Modality Hallucination

To hallucinate a modality, a projection will be used to transform our RGB features to our IMU features. A multi-layer perceptron will be used to complete this projection. The RGB-to-IMU projection model

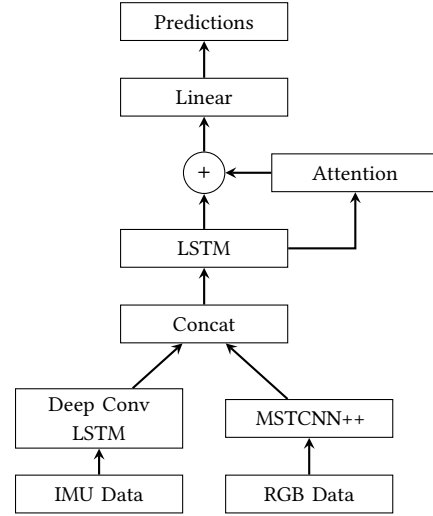


Figure 1: Architecture for our initial multi-modal system

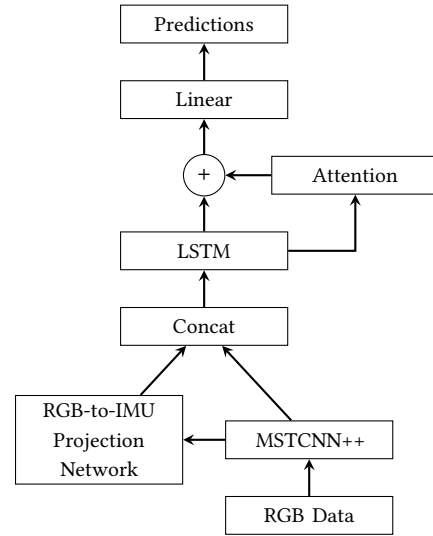


Figure 2: Hallucination technique in a multi-modal system

takes as input an RGB embedding and outputs a corresponding IMU embedding. This network is defined by its fully connected layers because we wanted to identify a stateless series of computations to transition from one modality to the other modality. This network is evaluated using the cosine similarity between the predicted IMU vector and the corresponding true IMU vector. After computing the projected the IMU vectors, the method from 3.4.1 will be used on the 50salads dataset and be compared with the other methods to see whether using the projected data will be effective in a multimodal setting.

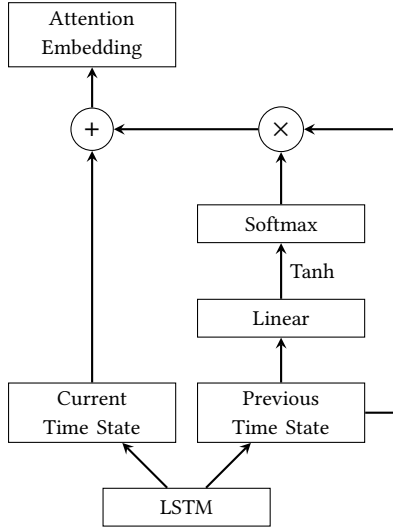


Figure 3: A diagram of the attention model used.

Modalities	Model	Accuracy
RGB	MLP	63.35%
RGB	MSTCN++	84.96%
IMU	MLP	48.99%
IMU	DeepConvLSTM	67.23%
RGB, IMU	MLP (-)	49.15%
RGB, IMU	Projection MLP (-)	48.92%
RGB, IMU	MLP	55.10%
RGB, IMU	Projection MLP	53.50%
RGB, IMU	LSTM (-)	54.11%
RGB, IMU	Projection LSTM (-)	53.25%
RGB, IMU	LSTM	57.50%
RGB, IMU	Projection LSTM	57.40%

Table 1: Our test accuracy scores on different architectures across modalities. For multimodal networks ending with (-), attention modules were not used. Otherwise, multimodal networks use attention modules and residual connections.

3.6 Experiments and Evaluation Metrics

To better quantify our experiments and assess our results, our main metric used will be accuracy to determine the impact of embedding hallucination. These results will be shown in the next section.

4 RESULTS AND DISCUSSION

4.1 Results

Table 1 lists the experiments attempted and the accuracies found from each. The experiments were performed on the 50salads dataset, as aforementioned. The dataset provides both RGB and IMU data to run experiments on. Table 2 shows experiments attempted and the macro F1 scores found from each. Table 3 shows cosine similarity results for the projection network used. Classification models were evaluated using cross entropy loss. We used an algorithm to

Modalities	Model	Macro F1
RGB	MLP	48.35%
RGB	MSTCN++	43.54%
IMU	MLP	39.78%
IMU	DeepConvLSTM	52.19%
RGB, IMU	MLP (-)	41.84%
RGB, IMU	Projection MLP (-)	40.90%
RGB, IMU	MLP	46.70%
RGB, IMU	Projection MLP	45.50%
RGB, IMU	LSTM (-)	44.14%
RGB, IMU	Projection LSTM (-)	42.55%
RGB, IMU	LSTM	49.40%
RGB, IMU	Projection LSTM	49.20%

Table 2: Our macro F1 scores on different architectures across modalities. For multimodal networks ending with (-), attention modules were not used. Otherwise, multimodal models use attention modules and residual connections.

Network	Cosine Similarity
RGB-to-IMU Projection	82.10%

Table 3: Our results for the projection network used.

compute the learning rate for each model. Our batch size was set to 32 and we used dropout layers for regularization.

4.2 Singular Modality

4.2.1 RGB. Comparing the RGB-only modality experiments, the MS-TCN++ does a much better job of classification than the multi-layer perceptron. As previously mentioned, although the complexity of the multi-layer perceptron (MLP) is elementary, the model’s learning suffers because of it. This can be seen in the table, where accuracy score is low at 63.35%. On the other hand, the MS-TCN++ model boasts a high accuracy, 84.96%, on the test 50salads dataset. MS-TCN++ has a more complex and sophisticated architecture, where it can capture temporal dependencies between action classes, reduce over-segmentation errors, and introduces a dual dilated layer that captures both local and global features [4]. The model is fully convolutional, it is very efficient and fast both during training and testing. From this analysis, it is shown that the MS-TCN++ would be best for the RGB modality.

4.2.2 IMU. Comparing the IMU-only modality experiments, the DeepConvLSTM does a much better job of classification than the multi-layer perceptron. As stated above, the multi-layer perceptron has a low complexity level, but suffers from a lack of learning from the data. The MLP only accurately identifies less than half of the actions, at a rate of 48%. The DeepConv LSTM proved to be better suited than the multi-layer perceptron to classify the IMU action data at a rate of 67% accuracy. With its convolutional and LSTM recurrent layers, the data can be viewed as a set of IMU data instead of individual points. In turn, a higher accuracy is expected and seen in the experiments. Additionally, from Table 2, it can be seen that DeepConvLSTM had the highest F1 score at 52.19%. This model has

the best harmonic mean between precision and recall, and it would be a

4.3 Multiple Modalities

4.3.1 Feature Concatenation and MLP. To use multiple modalities for modeling, the RGB and IMU features were concatenated before passing it as input into a model as discussed in 3.4.1 [9]. When combining both modalities, the accuracy with a multi-layer perceptron, 57.98%, is inferior to accuracy of the MS-TCN++ with the RGB data and the DeepConv LSTM with the IMU data. This suggests the ineffectiveness of using more sensor data, despite the assumption in activity recognition field that more data yields better results.

4.3.2 Projected IMU Data, Feature Concatenation and MLP. The projected network to generate IMU embeddings, the feature concatenation with the RGB data and the projected IMU embeddings, and then passing it as input into a multi-layer perceptron seems to do not as well as the method in 4.3.1. For instance, in Table 1, the projected multi-modal MLP with attention does slightly worse than the multi-modal MLP with attention, 46.70 % versus 45.50%. The multi-modal projected LSTM does slightly worse than the multi-modal LSTM, 49.40% versus 49.20%. This shows that the projected modality approach retains some performance even if the second modality is unavailable. In this case, we projected RGB embeddings into IMU embeddings, but this could be done for any pair of modalities. In addition, it may be possible to do apply this approach for more than two modalities and see success.

4.4 Attention Mechanisms

The proposed attention mechanisms were effective in their intended approach as seen in Table 2, resulting in higher accuracies for both multi-modal and uni-modal approaches. This is due to the attention mechanisms focusing on the correct parts of the input and weighting the features correctly.

5 CONCLUSION

Unfortunately, we were unable to construct a model which matched or outperformed the classification made by uni-modal constructions. The model that uses multiple modalities may not have been able to generalize beyond the training data provided. However, we have shown that more complicated multimodal networks perform better than the less complicated networks (MLP versus LSTM), so we believe that a more complicated network designed for multi-modal learning would lead to results which outperform uni-modal architectures. There are better networks that could be looked into for multi-modal learning and future work into this could be of benefit. Pre-processing of the data could also be looked into. Using different approaches for pre-processing could lead to better performance, along with higher performing fusion techniques.

All in all, the effectiveness of sensor fusion seem to be weak. This may be due to an imbalanced dataset, faulty sensor data, or even learning issues within the model that need to be determined. However, attention-enhanced and projection networks show promise in future networks. This experimentation showed the value of adopting convolutional-first technique approaches, as seen in the MS-TCN++. Along with the future work discussed, looking at convolutional-first techniques may provide a unique advantage.

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